# Circuit Lab, Division B

**Practice Test 3 (State/National level)** · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

**Instructions**

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

| Section | Questions |
|---|---|
| Section I: Multiple choice | 1–14 |
| Section II: Short answer and calculations | 15–40 |
| Section III: Circuit problems | 41–50 |

## Section I: Multiple choice

*Circle the one best answer. 1 point each.*

**1.** Kirchhoff first published his circuit laws, as a student, in which year?

**A.** 1785 **B.** 1827
**C.** 1845 **D.** 1887

**2.** Ohm's 1827 book *Die galvanische Kette, mathematisch bearbeitet* was attacked at first mainly because:

**A.** It claimed that current flows from negative to positive
**B.** It said that charge is quantized
**C.** Its numbers came from a torsion balance
**D.** It used abstract mathematics to describe a physical experiment

**3.** In 1887, while he studied sparks, Hertz noticed that ultraviolet light shining on a spark gap:

**A.** Stopped the spark completely
**B.** Changed the color of the spark to violet
**C.** Made the spark jump more easily
**D.** Reversed the direction of the current

**4.** Coulomb measured the inverse-square law for electric force in 1785 with which instrument?

**A.** A voltaic pile **B.** A torsion balance
**C.** A galvanometer **D.** A spark-gap oscillator

**5.** Static shocks from doorknobs are most common:

**A.** On humid summer days **B.** When it rains
**C.** When the doorknob is wet **D.** On dry winter days

**6.** The hot wire inside a metal toaster comes loose and touches the metal case. The case is connected to the ground wire. What happens?

**A.** Nothing; the toaster keeps working and the case is safe forever
**B.** The neutral wire carries the extra current, so the case becomes positive
**C.** The case stays at 120 V, but no current flows
**D.** A large current flows to ground and trips the breaker (or blows the fuse)

**7.** A meter reads 0.004050 A. How many significant figures does this reading have?

**A.** 2 **B.** 3
**C.** 4 **D.** 7

**8.** A hair dryer on a GFCI outlet draws 7.500 A in the hot wire, but only 7.491 A returns in the neutral wire. What does the GFCI do?

**A.** Nothing, because 7.5 A is below 15 A
**B.** It trips, because the current is above 7 A
**C.** Nothing, because the difference is less than 1%
**D.** It trips, because 9 mA leaks to ground

**9.** A lamp is wired with two 3-way switches. It is OFF when both switches are down (0, 0), and flipping either switch changes the lamp. Which logic function gives the lamp state?

**A.** AND **B.** OR

**C.** NAND **D.** XOR

**10.** A lamp circuit with cold tungsten filaments blows its fast-blow fuse each time it is switched on, but runs well below the fuse rating once warm. The best fix is:

**A.** A slow-blow (time-delay) fuse of the same rating

**B.** A fuse with a much higher rating

**C.** A copper wire in place of the fuse

**D.** A fast-blow fuse with a lower rating

**11.** What is the smallest number of 2-input NOR gates that makes a 2-input AND gate?

**A.** 1 **B.** 2

**C.** 3 **D.** 4

**12.** Simplify $F = ((A + B)' \cdot C)'$.

**A.** $A'B'C$ **B.** $(A + B)C'$

**C.** $A + B + C'$ **D.** $AB + C'$

**13.** Twelve 60 Ω resistors form the edges of a cube. What is the resistance between two corners that are at the opposite ends of a diagonal of ONE FACE?

**A.** 30 Ω **B.** 35 Ω

**C.** 45 Ω **D.** 50 Ω

**14.** A 9.0 V battery feeds one 360 Ω resistor. After the resistor, a red LED ($V_F$ = 1.8 V) and a blue LED ($V_F$ = 3.1 V) are connected in PARALLEL to the negative terminal. What happens?

**A.** Only the red LED lights; it gets about 20 mA

**B.** Both LEDs light; each gets about 10 mA

**C.** Only the blue LED lights; it gets about 16 mA

**D.** Neither LED lights

## Section II: Short answer and calculations

*Write the answer with units and the correct number of significant figures. 2 points each.*

**15.** Three point charges lie on the x-axis: +4.0 μC at x = 0, −2.0 μC at x = 0.30 m and +3.0 μC at x = 0.50 m. Find the net force on the −2.0 μC charge (size and direction).

**16.** A +4.0 μC charge is at x = 0 and a −1.0 μC charge is at x = 0.30 m. At what point on the x-axis (not at infinity) is the electric field zero?

**17.** Charge $q_1$ = +3.0 μC is at the origin. $q_2$ = −2.0 μC is at (0.30 m, 0) and $q_3$ = +5.0 μC is at (0, 0.40 m). Find the size of the net force on $q_1$. Also give its x and y parts with signs.

**18.** Two identical small metal spheres carry +8.0 μC and −2.0 μC. They touch and are then put 0.15 m apart. Find the new force between them. Attract or repel?

**19.** A battery has an EMF of 9.0 V. With a 10.0 Ω load, its terminal voltage is 8.2 V. Find the internal resistance of the battery.

**20.** For the same battery, what is the LARGEST power it can deliver to any load resistor?

**21.** A battery pack has 4 cells in series in each string and 2 strings in parallel. Each cell is 3.7 V, 2600 mAh. Find the pack voltage and the stored energy in watt-hours.

**22.** Mains power in Europe is 230 V rms. What is the peak-to-peak voltage?

**23.** A copper coil measures 5.20 Ω at 20 °C. When it is hot, it measures 6.10 Ω. Copper has α = 0.0039 per °C. Find the hot temperature.

**24.** A house replaces ten 60 W incandescent bulbs with ten 9.0 W LED bulbs. Each bulb is on 5.0 hours a day for 365 days. Electricity costs $0.15 per kWh. How much money is saved in the year?

**25.** An aluminum wire ($\rho = 2.65 \times 10^{-8}$ Ω·m) must have the SAME length and the SAME resistance as a copper wire ($\rho = 1.68 \times 10^{-8}$ Ω·m). How many times larger must its diameter be?

**26.** Two 100 kΩ resistors are in series across 12.0 V. A cheap voltmeter with an internal resistance of 200 kΩ is placed across the bottom resistor. What does it read? (Without the meter the voltage is 6.00 V.)

**27.** An infinite ladder network has a 20 Ω resistor in series in each step and a 60 Ω resistor across (shunt) in each step. The input starts with a series 20 Ω resistor. Find the input resistance.

**28.** A 9.0 V battery drives 220 Ω in series with a parallel pair: 330 Ω and an unknown R. The battery current is 25 mA. Find R.

**29.** Three bulbs rated 40 W, 60 W and 100 W (all rated for 120 V) are connected in SERIES across 120 V. Which bulb is brightest, and what power does it use? (Treat each bulb as a fixed resistor.)

**30.** Three identical bulbs: bulb 1 is in series with a parallel pair (bulbs 2 and 3), on an ideal battery. Bulb 3 is unscrewed. By what factor does the power of bulb 2 change?

**31.** A 1500 W heater on 120 V runs for many hours (a continuous load). Using the 80% rule, what is the smallest standard breaker (15, 20, 25 or 30 A) it needs?

**32.** A +12.0 V supply connects through a silicon diode (anode at the 12 V side, 0.7 V drop) and a 1.0 kΩ resistor to a +5.0 V supply. Both supplies share the same ground. Find the current.

**33.** A silicon diode has a forward voltage of 0.650 V at 25 °C. Its forward voltage drops about 2.0 mV per °C. Estimate the forward voltage at 85 °C (same current).

**34.** Use a K-map to find the simplest sum of products for F(A, B, C, D) = Σm(0, 2, 6, 8, 9, 12, 14) with don't cares d(1, 4, 10, 11).

**35.** A full adder has inputs A = 1, B = 0 and $C_{in} = 1$. Give the sum S and the carry-out $C_{out}$.

**36.** Three judges (A, B, C) each press a button (1 = yes). A light turns on when at least two judges press yes. Write the simplest Boolean expression for the light.

**37.** Add the binary numbers 10111011 + 1101101. Give the answer in binary and in decimal.

**38.** A 9.00 V battery is in series with a known 4.70 kΩ resistor and a mystery resistor. The only meter is a voltmeter. It reads 9.00 V across the battery and 3.15 V across the mystery resistor. Find the mystery resistance.

**39.** A tablet battery is 5000 mAh at 3.85 V. The fast charger label says 9.0 V, 2.0 A. The charging is 82% efficient. Estimate the time for a full charge from empty.

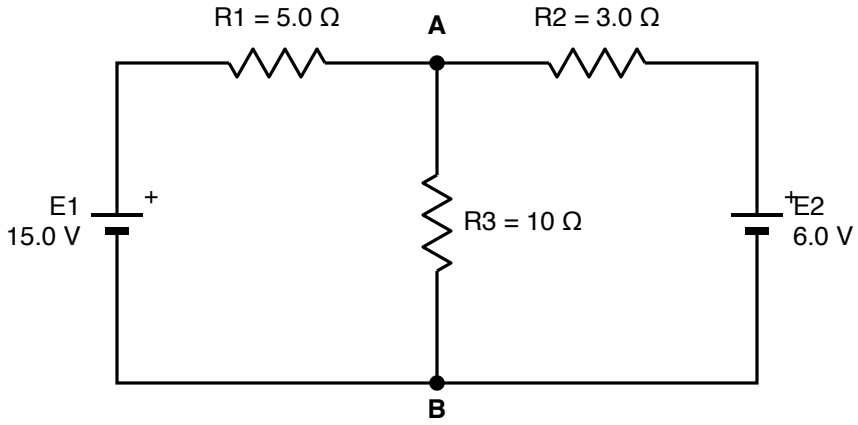
**40.** You have one each of 100 Ω, 220 Ω, 330 Ω and 470 Ω. Show how to make a resistance within 1% of 253 Ω, and give its exact value.

## Section III: Circuit problems

*Show your work. 2 points per part.*

**Problem A: two sources (Kirchhoff) (Questions 41–44)**

In the circuit below, E1 = 15.0 V and E2 = 6.0 V are ideal batteries (the + side is the long plate, at the top of each). R1 = 5.0 Ω, R2 = 3.0 Ω, R3 = 10 Ω. Let $I_1$ be the current in R1 flowing toward node A and $I_2$ the current in R2 flowing toward node A.




**41.** Write the two loop equations and find $I_1$.

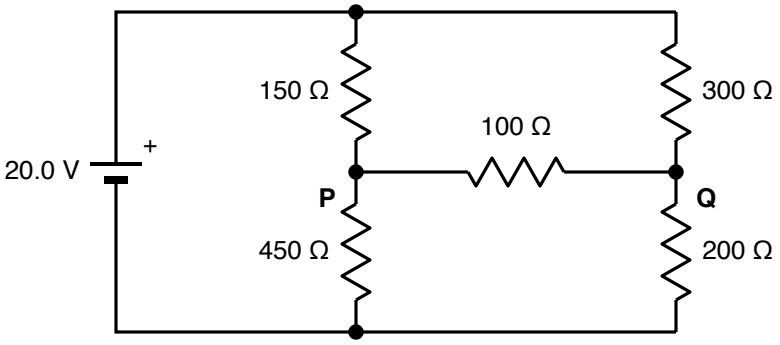
**42.** Find $I_2$. Which way does the current really flow through E2?

**43.** Find the voltage of node A relative to node B.

**44.** How much power does E2 absorb (take in)?

**Problem B: unbalanced bridge (Questions 45–46)**

A 20.0 V ideal battery feeds the bridge below. Node P is between 150 Ω and 450 Ω; node Q is between 300 Ω and 200 Ω. A 100 Ω resistor $R_{PQ}$ joins P and Q.

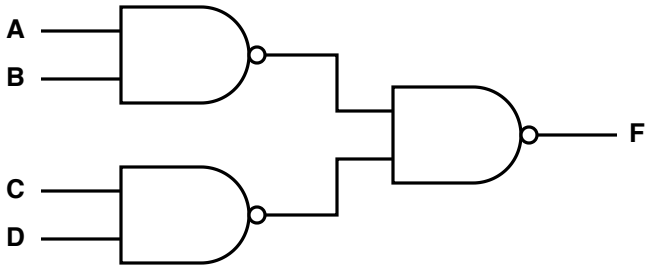



**45.** First, $R_{PQ}$ is removed. What would an ideal voltmeter read between P and Q ($V_P - V_Q$)?

**46.** Now $R_{PQ}$ is in place. Find the current in $R_{PQ}$ and its direction.

**Problem C: logic circuit (Questions 47–48)**

The circuit below has three NAND gates, inputs A, B, C, D and output F.

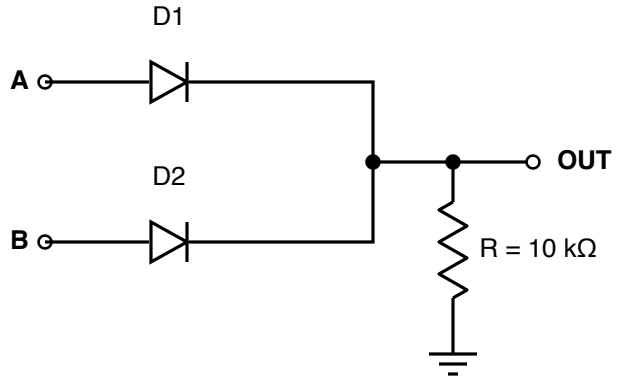



**47.** Write F as a simple sum of products.

**48.** For how many of the 16 input rows is F = 1?

**Problem D: diode logic (Questions 49–50)**

Inputs A and B are each either 0 V or 5.0 V. D1 and D2 are silicon diodes (0.7 V when on). R = 10 kΩ goes from the output to ground.




**49.** Which logic gate does this circuit make (positive logic: 5 V = 1)?

**50.** A = 5.0 V and B = 0 V. Find the output voltage and the current in R.

**End of the test. Check your answer sheet.**

## Answer sheet · Practice Test 3 (State/National level)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

**1.** A B C D

**2.** A B C D

**3.** A B C D

**4.** A B C D

**5.** A B C D

**6.** A B C D

**7.** A B C D

**8.** A B C D

**9.** A B C D

**10.** A B C D

**11.** A B C D

**12.** A B C D

**13.** A B C D

**14.** A B C D

**15.** _____

**16.** _____

**17.** _____

**18.** _____

**19.** _____

**20.** _____

**21.** _____

**22.** _____

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**47.** _____

**48.** _____

**49.** _____

**50.** _____